La Consolacion University Philippines
Valenzuela St., Capitol View Park Subdivision, Bulihan, City of Malolos, Bulacan

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SCHOOL YEAR 2025-2026
COLLEGE OF INFORMATION TECHNOLOGY AND ENGINEERING

I. COURSE CODE : CPE-404

II. COURSE TITLE : Digital Signal Processing

III. COURSE DESCRIPTION :

This course provides students with a comprehensive understanding of the fundamental concepts and practical applications of Digital Signal Processing (DSP) using modern computational tools such as MATLAB. Topics include discrete-time signals and systems, sampling, z-transforms, Fourier analysis, filter design, and signal reconstruction. Emphasis is placed on applying mathematical and engineering principles to real-world signal processing problems, including audio, image, and communication systems.

The course integrates Marian Augustinian values, fostering not only intellectual and technical competence but also ethical responsibility and social awareness. Students are encouraged to embody truth, unity, and charity through collaborative projects, demonstrating both technical proficiency and moral integrity as future Christo-centric Catholic professionals in the field of computer engineering.

IV. PRE-REQUISITE : CPE-303

V. CREDIT UNITS : 3 Units Lecture, **1** Laboratory

VI. COURSE LEARNING OUTCOMES:

Vision	Mission

Moved by the Spirit of the Risen Christ, La Consolacion University Philippines, envisions itself to become a catalyst for the wholistic transformational development of Catholic Augustinian Marian graduates rooted in Gospel values.	To achieve the above vision, LCUP commits itself to generate and make available to its Augustinian Marian students, programs and services that will make them: ● *exhibit academic and technical competence in their respective areas of discipline in the local and international environment;* ● *demonstrate their knowledge, critical thinking and creativity by exploring the various avenues of learning and searching for truth;* ● *live the Gospel values in the varied aspects of their personal and professional lives;* ● *manifest their willingness to serve specially the marginalized sector of our society;* ● *demonstrate productivity in various areas of learning as responsible citizens of the society; and* ● *develop individuals who are resilient, inclusive and responsive to be socio-economic realities of the Philippines and the complexities of the global environment.*

EXPECTED GRADUATE ATTRIBUTES	PROGRAM OUTCOMES	COURSE LEARNING OUTCOMES
EGA 1: Critical and Creative thinker EGA 2: Competent Catholic Augustinian-Marian Professional EGA 3: Socially Responsive Steward EGA 4: Transformative Lifelong learner	PO1. Ability to apply knowledge of mathematics and science to solve complex engineering problems; PO2. Ability to design and conduct experiments, as well as to analyze and interpret data; PO3. Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability and sustainability, in accordance with standards. PO4. Ability to function on	**CLO1:** identify the engineering fundamentals to analyze discrete-time signals and systems using computational tools. **[EGA1] [PO1, PO2, PO3, PO5]** **CLO2:** Demonstrate ethical responsibility and professionalism in designing DSP systems that uphold social, environmental, and moral standards.and communicate technical ideas effecfively **EGA1] [PO8, PO7, PO12]**

	multidisciplinary teams; PO5. Ability to identify; formulate, and solve complex engineering problems; PO6. Understanding of professional and ethical responsibility. PO7. Ability to communicate effectively; PO8. Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context. PO9. Recognition of the need for, and an ability to engage in life-long learning; PO10. Knowledge of contemporary issues; PO11. Ability to use techniques, skills and modern engineering tools necessary for engineering practice; and PO12. Knowledge and understanding of engineering and management principles as a member and leader in a team, to manage projects and in multidisciplinary environments. PO13: Ability to design, develop, implement and improve integrated system that include people materials, implementation, equipment and energy.	**CLO3:** Evaluate the societal and environmental impacts of DSP technologies, ensuring responsible innovation in engineering solutions. **[EGA3] [PO8, PO10]** **CLO4:** Engage in continuous learning and exploration of emerging DSP techniques and tools to adapt to technological advancements. **[EGA4] [PO 9, PO11]**

VII. LEARNING PLAN:

SCHEDULE	INTENDED LEARNING OUTCOMES	COURSE CONTENTS	SUGGESTED LEARNING ACTIVITIES	SUGGESTED ASSESSMENT TASKS	EXPECTED VALUE/S TO BE INTEGRATED
Week 1		**I.Orientation** • Introduction to the Course Contents • Course Requirements and Activities • Course Objectives • Grading System and Policies on Absences, Tardiness and Examinations	• Setting and sharing of Expectations from the Course • Discussion on the Vision, Mission and Goals of LCUP and of CITE • Presentation of Course Objectives and Contents, Course Requirements and Activities Review of Policies on Absences, Tardiness, Examinations, Grading System	• Statement of Expectations • Review of Course Contents, Objectives, Activities and Assessment Tools • Review of Policies on Absences, Tardiness, Examinations, Grading System	Understanding Open mindedness Respect Obedience
Week 2-3	ILO1: Show understanding, and deep appreciation on the nature and components of digital signal processing. [CLO1] ILO2: Show understanding in the fundamental concepts of digital signal processing. [CLO1]	**II.Introduction to Digital Signal Processing** • What is signal processing • Need of processing • Categories • Advantages of digital over analog • Filters-analog & digital • Comparison	Lecture-Discussion using Power Point presentation	• Quiz on the topics discussed Activity on the application of DSP	Cooperation Understanding Active participation and involvement Open mindedness

	ILO3: Show understanding in practical appli cation of DSP. [CLO3][CLO4] ILO4: Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing. [CLO2]	• Digital filters-types •			dence in answering and participating in classroom discussion
Week 4-5	ILO1: Show understanding, and deep appreciation on the nature and components of digital signal processing. [CLO2] ILO2: Show understanding in the fundamental concepts of digital signal processing. [CLO2] ILO4: Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing. [CLO2]	**III. Application of Digital Signal Processing** • FIR & IIR • Advantages-disadvantages • Applications of dsp • Simple illustrations • Various types of DSP • Microprocessor & signal processor • Architecture • Integrators • Companding • Bandwidth • Restriction • Sensors • Transducers •	Lecture-Discussion using Power Point Presentation	Quiz on the topics discussed	• Patience • Understandin g • Resourcefuln ess • Open mindedness Active participation and involvement

Week 6		• **Preliminary Examination**		Preliminary Examination	• Understanding • Honesty Problem-Solving
WEEK 7-8	ILO1: Show understanding, and deep appreciation on the nature and components of digital signal processing. [CLO1] ILO2: Show understanding in the fundamental concepts of digital signal processing. [CLO1] ILO4: Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing. [CLO2]	**IV. Converting Analog to Digital Signals** • Block diagram of DSP • Digital signal processor • Systems that process digital signals • Application of DSP • Sampling • Sampling interval • Short hand notation forsamples • Ways to represent sample signals • Sampling instant •	• Lecture-Discussion using Power Point Presentation	Quiz on the topics discussed	• Patience • Understanding • Resourcefulness • Open mindedness Active participation and involvement
Week 9-11	ILO1: Show understanding, and deep appreciation on the nature and components of digital signal processing. [CLO1] ILO2: Show understanding in the fundamental concepts of digital signal processing. [CLO1]	**V. Introduction to MATLAB** • MATLAB Command Window • Arithmetic Operations with Scalars Operation Symbol • Order of Precedence Parentheses	• Lecture-Discussion using PowerPoint Presentation • Discovery Learning Problem solving	• Recitation Seatwork	• Patience • Understanding • Resourcefulness • Open mindedness • Active participation and involvement

	ILO3: Show understanding in practical application of DSP. [CLO3][CLO4] ILO4: Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing [CLO3]	Exponentiation Multiplication • Display Formats • Command Description format • Math functions - Function Description • Trigonometric math functions • Defining scalar variables • Useful MATLAB commands • Rules about variable names • Predefined variables • Array List of numbers arranged in row and/or columns • Creating vector Creating 2D array(matrix)			
Week 12		**Midterm Examination**		Midterm Examination	• Understanding • Honesty Problem-Solving
Week 13	ILO2: Show understanding in the fundamental concepts of digital signal processing. [CLO1]	**VI. Correlation** • Definition • Autocorrelation • Cross correlation • 3 operation of correlation	• Lecture-Discussion • Discovery Learning	• Recitation • Quiz Seatwork	• Patience • Understanding • Resourcefulness

	ILO3: Show understanding in practical application of DSP. [CLO3] [CLO4] ILO4: Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing. [CLO2]	• Calculating and graphing of cross-correlation between two signals • Calculating and graphing of autocorrelation • Conclusion on correlation • Deterministic signals Random noise			• Open mindedness Active participation and involvement
Week 14	ILO2: Show understanding in the fundamental concepts of digital signal processing. [CLO1] ILO4: Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing. [CLO2]	**VII. Decibel Unit** • The Bel unit • Computation of dB • Quantization Error • Application of digital system in accordance to the number of bits • Tolerance of Decibel in human hearing • Aliasing • Anti-aliasing filter • Amplifiers • Filter	• Lecture-Discussion Problem-Solving	• Recitation Quiz	• Patience • Understanding • Resourcefulness • Open mindedness Active participation and involvement
Week 15	ILO2: Show understanding in the fundamental concepts of digital signal processing. [CLO1] ILO3: Show understanding in	**VIII. Convolution** • Definition • Flipping • Formula for convolution • Calculating and graphing of convolution	• Lecture-Discussion Problem-Solving	• Recitation Quiz	• Patience • Understanding • Resourcefulness • Open mindedness

	practical appli cation of DSP. [CLO3][CLO4] ILO4: Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing. [CLO2]	• Application of convolution			Active participation and involvement
Week 16-17	ILO2: Show understanding in the fundamental concepts of digital signal processing. [CLO1] ILO3: Show understanding in practical appli cation of DSP. [CLO3][CLO4] ILO4: Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing. [CLO2]	**IX. Periodic Functions** • Definition • Relationship between period and frequency • Prime example of periodic functions • Sinusoidal wave • Amplitude of sinusoidal wave • Fundamental and Nth harmonics • Time domain and frequency domain Constructing a waveform with sine waves	• Lecture-Discussion • Presentation of different waves in MATLAB Problem Solving	• Recitation • Quiz Seatwork	• Patience • Understanding • Resourcefulness • Open mindedness Active participation and involvement
Week 18		**Final Examination**		Final Examination	• Understanding • Honesty • Problem-Solving

VIII. COURSE REQUIREMENTS :

1. Attendance
2. Oral presentation of assigned topic
3. Participation in class
4. Project Submission
5. Periodic written examination

IX. REFERENCES:

A. INTERNET REFERENCES

Proakis, J. G., & Manolakis, D. G. (2021). *Digital signal processing: Principles, algorithms, and applications* (5th ed.). Pearson.

B. **JOURNALS:**

Zou, C., & Wang, S. (2024). Robust quaternion elastic-net for hypercomplex signal recovery. *Digital Signal Processing*. ACM Digital Library

Niu, Z. (2024). Learnable digital signal processing: A new benchmark of performance. *Nature Photonics*. Nature

Gomes de Sousa, D., Alves, C. S. M., da Silva, V. L., & Ferreira Neto, N. A. (2025). Digital Signal Processing from Classical Coherent Systems to Continuous-Variable QKD: A Review of Cross-Domain Techniques, Applications, and Challenges. *arXiv*. arXiv

Hayes, B., Shier, J., Fazekas, G., McPherson, A., & Saitis, C. (2023). A Review of Differentiable Digital Signal Processing for Music & Speech Synthesis. *arXiv*. arXiv

Muda, L., Begam, M., & Elamvazuthi, I. (2010). Voice Recognition Algorithms using Mel Frequency Cepstral Coefficient (MFCC) and Dynamic Time Warping (DTW) Techniques. *arXiv*.

X. GRADING SYSTEM :

1. Class Standing 60%
   - Quizzes/Recitation
   - Group Participation
   - Seat Work
2. Examination 40 %
3. Term Grade CS [60%) + TE [40%)
4. Final Rating <u>Prelim Grade + Midterm Grade + Final Grade</u>

Prepared by:	Checked and Reviewed by:	Approved by:
ENGR. GLAZEN CAREY M. PEREZ Faculty/Date	**EDNA F. DAYAO, DIT** Dean/Date **ROSALYN S. GALVEZ, PhD., LPT** Asst. Vice President for Academic Affairs-Date References Certified by: **MARIA MERCEDEZ M. HAZ, PhD., RL** LIC Director/Date	**JAY MARK B. SANTOS, PhD, LPT** Vice President for Academic Affairs-Date

Revision Date:	
Modifications: __Learning Outcomes __Course Contents __Suggested Learning Activities __Suggested Assessment __Expected Values to be Integrated __Course Requirements __References __Grading System	